

insignificant whitespace. Keywords are words that are defined in the SQL language. They are either reserved (e.g., SELECT, COUNT and YEAR), or non-reserved (e.g., ASC, DOMAIN and KEY). Identifiers are names of database objects, like tables, columns and schemas. An identifier may not be equal to a reserved keyword, unless it is a delimited identifier (identifiers enclosed in double quotation marks) and can contain characters normally not supported in SQL identifiers. An identifier can also be identical to a reserved word, e.g., 'COUNT'.

Clauses are constituent components of statements and queries. Expressions can update values or change columns and rows of data in a table. Predicates are used for conditional statements to evaluate three-value logic (true/false/unknown) or Boolean truth values. Queries help retrieve the data based on specific criteria and it is an important element of SQL. Statements are used to do many database operations like transactions, database automation and diagnostics. Insignificant whitespace is generally ignored in SQL statements and queries, making it easier to format SQL code for readability.

SQL can be used to perform various logical operations on a database. For example:

```
CASE n WHEN 0
      THEN 'Zero'
      WHEN 1
      THEN 'One'
      ELSE 'Out of range!'
END
```

The above SQL 'CASE' statement is useful for conditional database operations.

Queries are important operations of SQL for any database. For example:

```
SELECT Car.brand AS Brand,
       count(*) AS Company
FROM Car
JOIN Car_company
```

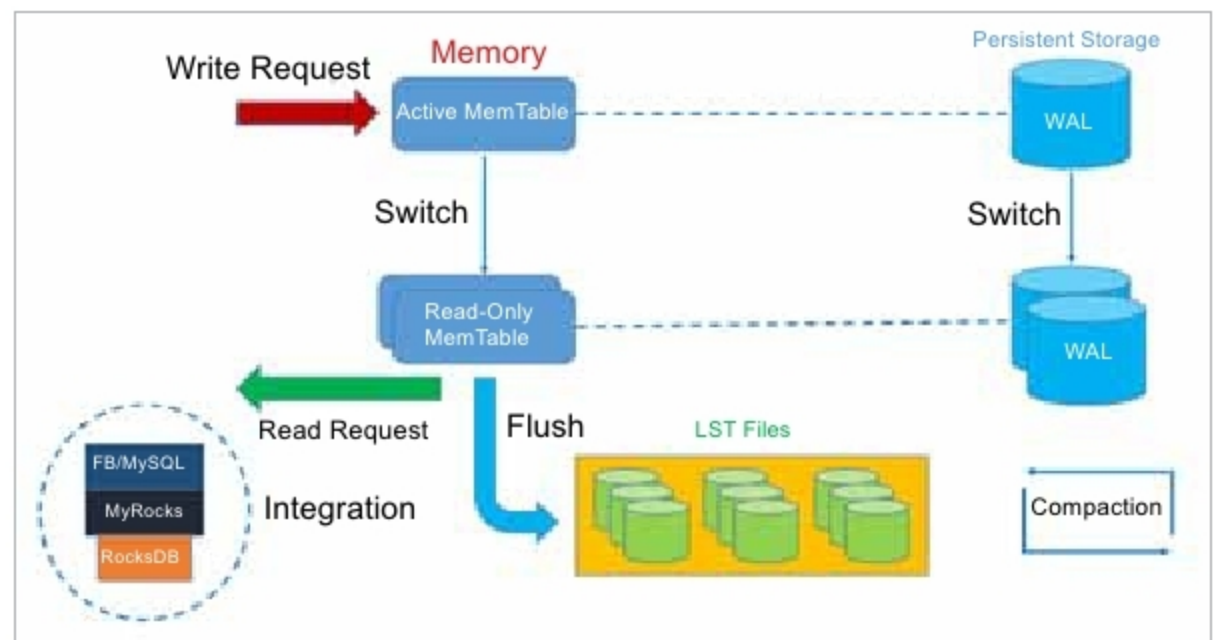


Figure 3: MyRocks architecture

```
ON Car.price = Car_company.price
GROUP BY Brand;
```

The above statement uses the 'SELECT' clause to retrieve data about cars from an inventory database of sellers, based on the price and brand.

As SQL operations are important for database operations, MyRocks is used with RocksDB to support MySQL. MyRocks integrates RocksDB with MySQL and helps to run SQL commands. MyRocks is a database engine that stores data using RocksDB as Static Sorted Table (SST) files in the SSDs. Such files are segmented using the Log Structured Merge Tree (LSMT) concept and stored in the disk.

Any write request made to the memory is stored in the Active Memtable and another copy is created as a Read-Only Memtable (ROM). When the data exceeds the memory, it is flushed to the disk as SST files using the LSMT concept. Simultaneously, log files called Write Ahead Log (WAL) are created and stored separately on the disk. These log files are later used to restore any data in case of a system failure. During a read request, the ROM is used from the memory to retrieve

data. MyRocks integrates RocksDB and MySQL to perform these operations flawlessly, using SQL.

## Features

MyRocks makes it possible to run MySQL with RocksDB, and has the following features.

- Greater space efficiency: MyRocks has 2x better compression compared to compressed InnoDB and 3x - 4x better compression compared to uncompressed InnoDB. This makes it possible to use less space.
- Greater writing efficiency: MyRocks has a 10x less write amplification compared to InnoDB and gives better endurance of Flash storage, which improves the overall throughput.
- Faster replication: No random reads for updating secondary keys, except for unique indexes. The *Read-Free Replication* option restricts any random reads when updating primary keys and this makes faster replication.
- Faster data loading: MyRocks writes data directly onto the bottom-most level and avoids all compaction overheads when you enable faster data loading for a session. **END** 🐧

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